# Linear First-Order Equations & the Integrating Factor

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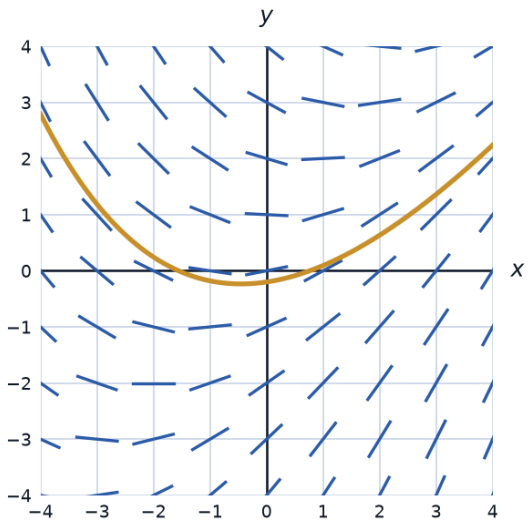
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Standard Linear Form and Integrating Factor

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

**2** [Foundational] • Target: Solving Linear First-Order IVPs

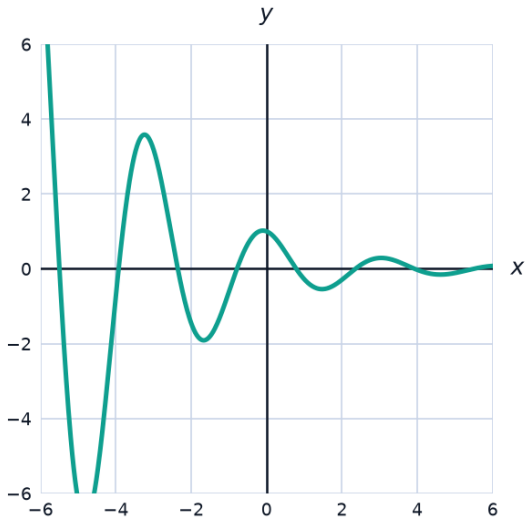
Consider the system of linear first-order differential equations given by: $(x'; y') = (1, 2; 3, 0)(x; y)$. Find the eigenvalues of the coefficient matrix.




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**3** [Foundational] • Target: Mixing and Tank Applications

Solve the first-order linear initial value problem $x\frac{dy}{dx} + 3y = \frac{\sin x}{x^2}$ for $x > 0$ with $y(\pi) = 0$. Find $\lim_{x \to 0^+} y(x)$.




**4** [Foundational] • Target: Transient and Steady-State Solutions

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter $k$ in $y_2(x) = kxe^x$.

**5** [Foundational] • Target: Standard Linear Form and Integrating Factor

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

**6** [Foundational] • Target: Solving Linear First-Order IVPs

Consider the first-order linear differential equation $\frac{dy}{dx} + 2y = 4$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

**7** [Foundational] • Target: Mixing and Tank Applications

Consider the system of linear differential equations: $\frac{dx}{dt}, = ax + by; \frac{dy}{dt}, = cx + dy$ where $a, b, c, d$ are constants. The characteristic equation for the system is $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$. We are given that the eigenvalues of the system are $\lambda_1 = 2$ and $\lambda_2 = -1$. The trace of the coefficient matrix, $T = a + d$, is $1$. Determine the value of the determinant of the coefficient matrix, $D = ad - bc$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** **[Intermediate]** • **Target: Transient and Steady-State Solutions**

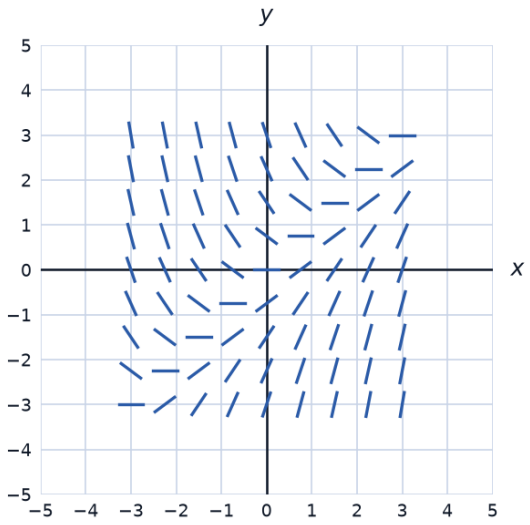
Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x \sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

**9** **[Intermediate]** • **Target: Standard Linear Form and Integrating Factor**

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1 e^{2x} + c_2 e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant $A$ such that $y_p(x) = Ax + B$ for some constant $B$.

**10** **[Intermediate]** • **Target: Solving Linear First-Order IVPs**

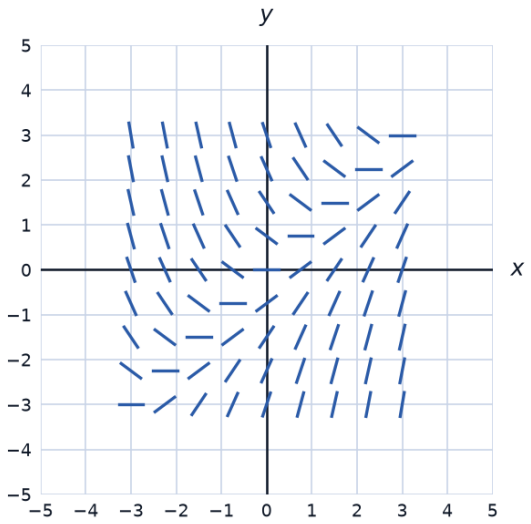
Find all equilibrium solutions of $\frac{dy}{dt} = y^2 - 4$, classify each as an attractor (sink) or repeller (source), and determine $\lim_{t \to \infty} y(t)$ for the solution with $y(0) = 0$.




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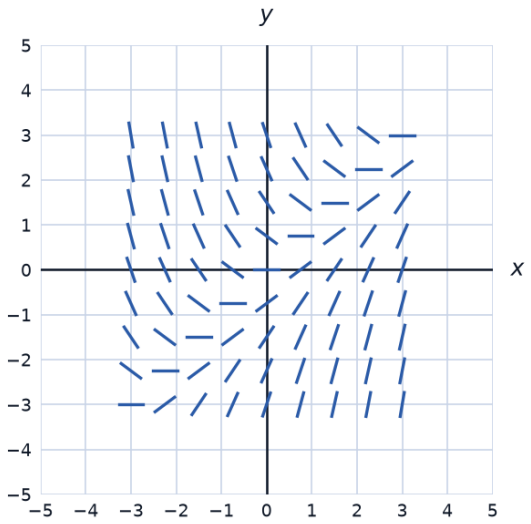
**11** [Intermediate] • Target: Mixing and Tank Applications

Solve the initial value problem $\frac{dy}{dx} = 2xy^2$ with $y(0) = 1$. Find $y(1/2)$ and state its maximal interval of valid existence.




**12** [Intermediate] • Target: Transient and Steady-State Solutions

Solve the second-order homogeneous initial value problem $y'' - 7y' + 10y = 0$ with $y(0) = 1$ and $y'(0) = -1$. Find $y(1)$.




**13** [Intermediate] • Target: Standard Linear Form and Integrating Factor

For the differential equation with a regular singular point at $x_0 = 0$: $2x^2y'' + xy' - (x+1)y = 0,$ find the indicial equation, determine the two roots $r_1 \geq r_2$, and find the first three non-zero terms of the Frobenius series solution corresponding to $r_1$.

**14** [Intermediate] • Target: Solving Linear First-Order IVPs

Solve the defective linear system $\mathbf{x}' = (1, -1; 1, 3)\mathbf{x}$ with $\mathbf{x}(0) = (2; 1)$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Mixing and Tank Applications** • **Cross-Lesson: 5.01 Variation of Parameters**

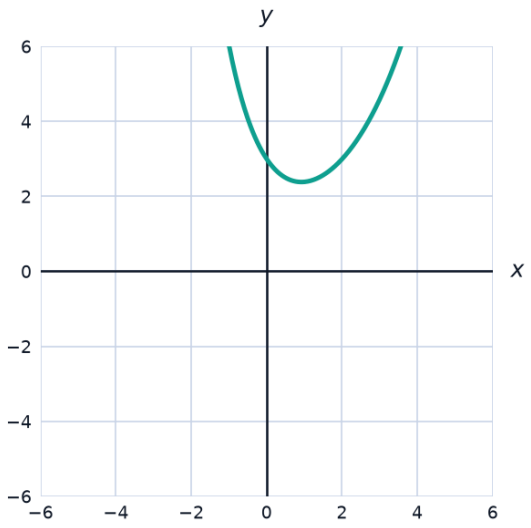
Find the power series recurrence relation about $x_0 = 0$ for Hermite's differential equation with parameter 2: $y'' - 2xy' + 4y = 0, \qquad y(0) = 1, \quad y'(0) = 0.$ Prove that the solution terminates to a quadratic polynomial and state the polynomial.

**16** **[Challenging]** • **Target: Transient and Steady-State Solutions** • **Cross-Lesson: 1.01 Separable Equations**

Find the general solution of the linear system $\mathbf{x}' = (1, 3; 3, 1)\mathbf{x}$. Classify the equilibrium at the origin and determine the cartesian equation of the stable separatrix line.

**17** **[Challenging]** • **Target: Standard Linear Form and Integrating Factor** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

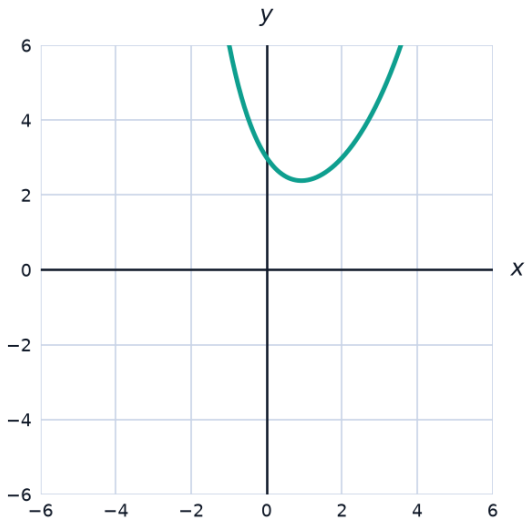
Solve the nonlinear Bernoulli differential equation $\frac{dy}{dx} - \frac{1}{x}y = 2x^2y^3$ for $x > 0$ with $y(1) = 1$. Find $y(2)$.




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**18** [Challenging] • Target: Solving Linear First-Order IVPs • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

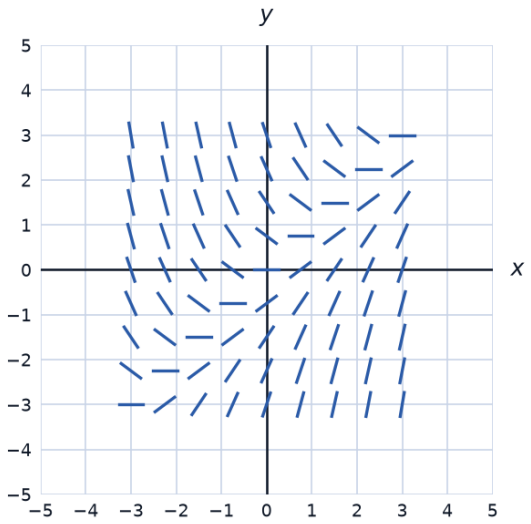
Solve the Cauchy-Euler differential equation $x^2y'' - xy' + 5y = 0$ for $x > 0$ with $y(1) = 2$ and $y'(1) = 4$.




**19** [Challenging] • Target: Mixing and Tank Applications • Cross-Lesson: 5.01 Variation of Parameters

Solve the undamped resonant initial value problem $y'' + 9y = 6\cos(3t)$ with $y(0) = 0$ and $y'(0) = 1$.

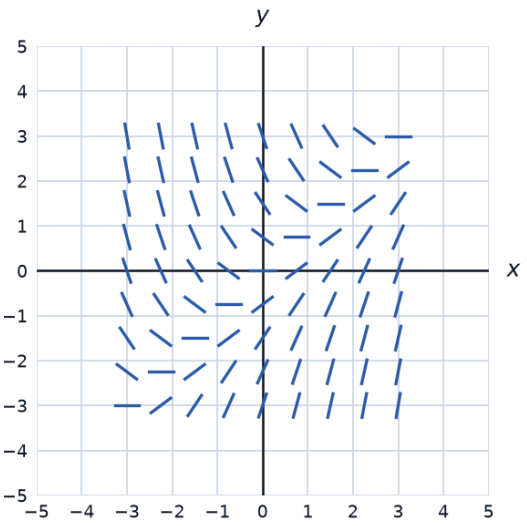
Identify the secular growth term.




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**20** [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

Find the general solution of $y'' - 4y' + 3y = 2x + e^{2x}$.




## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Standard Linear Form and Integrating Factor

Write the differential equation $x\frac{dy}{dx} + 2y = 4x^2$ in standard form $\frac{dy}{dx} + P(x)y = Q(x)$ for $x > 0$.

**a.** $\frac{dy}{dx} + \frac{2}{x}y = 4x$

**b.** $\frac{dy}{dx} + 2y = 4x$

**c.** $\frac{dy}{dx} + \frac{2}{x}y = 4x^2$

**d.** $\frac{dy}{dx} = 4x - 2y$

**22** [Foundational] • Target: Standard Linear Form and Integrating Factor

Compute the integrating factor $I(x) = e^{\int P(x)dx}$ for $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

**a.** $x^2$

**b.** $2\ln x$

**c.** $e^{2x}$

**d.** $x$

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**23** [Foundational] • Target: Standard Linear Form and Integrating Factor

Multiply the standard ODE by $I(x) = x^2$ to solve $\frac{d}{dx}[x^2 y] = 4x^3$.

**a.** $y = x^2 + \frac{C}{x^2}$

**b.** $y = 4x^2 + C$

**c.** $y = x^4 + C$

**d.** $y = x^2 + C$

**24** [Foundational] • Target: Solving Linear First-Order IVPs

Solve the IVP $\frac{dy}{dx} + y = e^{2x}$ with $y(0) = 1$.

**a.** $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$

**b.** $y = e^{2x}$

**c.** $y = \frac{1}{3}e^{2x} + e^{-x}$

**d.** $y = \frac{1}{2}e^{2x} + \frac{1}{2}e^{-x}$

**25** [Foundational] • Target: Solving Linear First-Order IVPs

Solve $\frac{dy}{dx} - 3y = 0$ with $y(0) = 4$.

**a.** $y = 4e^{3x}$

**b.** $y = 4e^{-3x}$

**c.** $y = 3e^{4x}$

**d.** $y = 4 + 3x$

**26** [Foundational] • Target: Solving Linear First-Order IVPs

What is the domain of validity for the solution to $x\frac{dy}{dx} + y = 1$ with $y(1) = 2$?

**a.** $(0, \infty)$

**b.** $(-\infty, \infty)$

**c.** $[1, \infty)$

**d.** $(-\infty, 0)$

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**27** [Foundational] • Target: Mixing and Tank Applications

A tank contains $100$ L of water and $10$ kg of dissolved salt. Pure water flows in at $3$ L/min, and the stirred mixture drains at $3$ L/min. What is the differential equation for the salt amount $A(t)$ in kg?

**a.** $\frac{dA}{dt} = -\frac{3}{100}A$

**b.** $\frac{dA}{dt} = 3 - \frac{3}{100}A$

**c.** $\frac{dA}{dt} = -3A$

**d.** $\frac{dA}{dt} = 10 - \frac{3}{100}A$

**28** [Foundational] • Target: Mixing and Tank Applications

For the tank above, what is the amount of salt remaining after $t$ minutes?

**a.** $A(t) = 10e^{-0.03t}$

**b.** $A(t) = 10 - 0.03t$

**c.** $A(t) = 10e^{-3t}$

**d.** $A(t) = 10e^{0.03t}$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Mixing and Tank Applications

If salt solution enters at $2$ kg/L at a rate of $3$ L/min, what is the inflow rate of salt in kg/min?

**a.** $6$ kg/min

**b.** $2$ kg/min

**c.** $3$ kg/min

**d.** $1.5$ kg/min

**30** [Intermediate] • Target: Transient and Steady-State Solutions

In the solution $y(t) = 5 + 4e^{-2t}$, which part represents the transient response and which represents the steady-state?

**a.** $5$ is steady-state, $4e^{-2t}$ is transient

**b.** $4e^{-2t}$ is steady-state, $5$ is transient

**c.** Both are steady-state

**d.** The entire response is transient

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**31** [Intermediate] • Target: Transient and Steady-State Solutions

What is the time constant $\tau$ for the decaying exponential $e^{-t/\tau} = e^{-5t}$?

**a.** $\tau = 0.2$ seconds

**b.** $\tau = 5$ seconds

**c.** $\tau = 1$ second

**d.** $\tau = 2.5$ seconds

**32** [Intermediate] • Target: Transient and Steady-State Solutions

After how many time constants ($t = n\tau$) has a decaying transient decayed to less than $1\%$ of its initial value?

**a.** $5$ time constants ($e^{-5} \approx 0.0067$)

**b.** $1$ time constant

**c.** $3$ time constants

**d.** $10$ time constants

**33** [Intermediate] • Target: Standard Linear Form and Integrating Factor

Consider the first-order linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$. If the integrating factor for this equation is given by $e^{x^2}$ and $Q(x) = 2xe^{x^2}$, find the value of the constant $C$ in the general solution if $y(0) = 3$. The general solution is of the form $y = e^{-x^2}\left(\int Q(x)e^{-x^2}dx + C\right)$.

**a.** 1

**b.** 3

**c.** 4

**d.** 6

**34** [Intermediate] • Target: Solving Linear First-Order IVPs

An RC circuit with $R = 4\,\Omega$ and $C = 0.5$ F is connected to $E(t) = 20\cos t$ V. What is the amplitude of the steady-state charge $q_{ss}(t)$?

**a.** $4\sqrt{5}$ C

**b.** $5$ C

**c.** $\sqrt{10}$ C

**d.** $2\sqrt{5}$ C

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**35** [Intermediate] • Target: Mixing and Tank Applications

An RL circuit has inductance $L = 5$ H and resistance $R = 15\,\Omega$. At $t = 0$, $i(0) = 0$ A. A constant voltage $E = 60$ V is applied. At what time $t$ does the current reach 90% of its steady-state value?

**a.** $\frac{1}{5}\ln 10$

**b.** $3\ln 10$

**c.** $\frac{1}{3}\ln(10/9)$

**d.** $\frac{1}{3}\ln 10$

**36** [Intermediate] • Target: Transient and Steady-State Solutions

An RL circuit with $L = 1$ H and $R = 2\,\Omega$ has initial current $i(0) = 0$ A. The applied voltage is $E(t) = 10\sin(2t)$ V. What is the amplitude of the steady-state alternating current $i_{ss}(t)$?

**a.** $5\sqrt{2}$ A

**b.** $\frac{5}{2}$ A

**c.** $\frac{5\sqrt{2}}{2}$ A

**d.** 5 A

**37** [Intermediate] • Target: Standard Linear Form and Integrating Factor

An RL circuit has $L = 3$ H, $R = 9\,\Omega$, and electromotive force $E(t) = 18\cos(3t)$ V. What is the steady-state current $i_{ss}(t)$?

**a.** $\cos(3t) - \sin(3t)$

**b.** $\cos(3t) + \sin(3t)$

**c.** $2\cos(3t) + 2\sin(3t)$

**d.** $\frac{1}{2}(\cos 3t + \sin 3t)$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Solving Linear First-Order IVPs • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A mixing tank contains 300 liters of brine with 15 kg of salt. Brine containing 0.1 kg/L of salt flows in at 6 L/min, and the mixture drains out at 6 L/min. What is the salt concentration in the tank as $t \to \infty$?

**a.** 0.05 kg/L

**b.** 0.1 kg/L

**c.** 0.2 kg/L

**d.** 0.15 kg/L

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**39** [Challenging] • Target: Mixing and Tank Applications • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation $y'' - 2y' + 5y = 0$. Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions. Suppose there exists a solution of the form $y(x) = c_1y_1(x) + c_2y_2(x)$ such that $y(0) = 1$ and $y'(0) = 2$. Furthermore, let the Wronskian of $y_1$ and $y_2$ be given by $W(y_1, y_2)(x) = e^{2x}$. Determine the value of the expression $y''(0) - 2y'(0) + 5y(0)$.

**a.** 0

**b.** $e^2$

**c.** 5

**d.** $2e^2$

**40** [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

A 120-gallon tank initially holds 60 gallons of brine containing 30 lb of dissolved salt. Brine containing 1 lb/gal enters at 3 gal/min, while the well-stirred mixture leaves at 1 gal/min. How much salt is in the tank at the exact instant the tank becomes full?

**a.** 90 lb

**b.** $120 - 15\sqrt{2}$ lb

**c.** $120 - 30\sqrt{2}$ lb

**d.** $120 - 15\sqrt{3}$ lb

**41** [Challenging] • Target: Standard Linear Form and Integrating Factor • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A 500-gallon tank initially contains 200 gallons of brine with 40 lb of salt. Brine containing 0.5 lb/gal of salt enters at 6 gal/min, and the well-stirred mixture drains at 4 gal/min. What is the salt concentration in lb/gal when the tank contains 300 gallons?

**a.** $\frac{37}{45}$ lb/gal

**b.** $\frac{41}{90}$ lb/gal

**c.** $\frac{1}{2}$ lb/gal

**d.** $\frac{37}{90}$ lb/gal

**42** [Challenging] • Target: Solving Linear First-Order IVPs • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank contains 100 L of water in which 20 kg of salt is dissolved. Brine containing 0.3 kg/L of salt enters at 5 L/min, and the mixture leaves at 3 L/min. What is the mass of salt in the tank after 50 minutes?

**a.** $60 - \frac{5\sqrt{2}}{2}$ kg

**b.** $60 - 5\sqrt{2}$ kg

**c.** $60 - \frac{5}{2}$ kg

**d.** $30 - \frac{5\sqrt{2}}{2}$ kg

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**43** [Challenging] • Target: Mixing and Tank Applications • Cross-Lesson: 5.01 Variation of Parameters

A tank initially holds 100 gal of brine containing 20 lb of salt. Brine containing 1 lb/gal enters at 4 gal/min, and the well-stirred mixture drains out at 6 gal/min. How much salt is in the tank after 25 minutes?

**a.** 45 lb

**b.** 40 lb

**c.** 35 lb

**d.** 50 lb

**44** [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

A tank initially contains 100 L of pure water. Brine containing 2 kg/L of salt flows in at 4 L/min, and the thoroughly mixed liquid flows out at 2 L/min. What is the total mass of salt in the tank after 50 minutes?

**a.** 300 kg

**b.** 250 kg

**c.** 350 kg

**d.** 200 kg

**45** [Challenging] • Target: Standard Linear Form and Integrating Factor • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A tank initially contains 80 L of brine containing 16 kg of salt. Brine containing 0.5 kg/L enters at 6 L/min, and the mixture flows out at 8 L/min. What is the amount of salt in the tank after 20 minutes?

**a.** $\frac{37}{2}$ kg

**b.** 19 kg

**c.** 18 kg

**d.** $\frac{35}{2}$ kg

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